MIL-C-45195E(AR)
25 March 1985
SUPERSEDING
MIL-C-45195D(PA)
7 June 1974

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE, M1
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of HE, Dualgran Cartridge designated as M1 for use with the 105MM Howitzer M2A2, M103 and M137.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-46565	- Primer Percusion, M28A2 and M28B2, Loading, Assembling and Packing.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by attributes.
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

MIL-STD-453 - Inspection, Radiographic.
MIL-STD-1235 - Single and Multi-level Continuous Sampling Procedures and Table for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

7549072 - Box, Packing, Ammunition for 105MM Howitzer Ammunition in Fiber Container, M105A3.
7549073 - Container, Ammo, Fiber, M105A3 for 105MM Howitzer Ammunition.
9211610 - Projectile, 105MM, HE, M1, Loading Assembly.
9211611 - Cartridge, 105MM, HE, M1, w/o Fuze for Howitzer M2A1, M2A2, M103 and M137 Assembly.
9216724 - Cartridge 105MM, HE, M1 w/Fuze PD, M557, for Howitzer M2A1, M2A2, M103 and M137 Assembly.
9212653 - Cartridge, 105MM, HE, M1 w/Fuze, for Howitzer M2A1, M2A2, M103 and M137 Assembly.
9258026 - Container, Ammo, Fiber, PA55, for 105MM Howitzer Ammunition Without Fuze.
9258027 - Box, Packing, Ammo for 105MM Howitzer, Ammo, w/o Fuze in Fiber Container PA 55.
9271850 - Packing and Marking for Box, Wirebound, Packing Ammunition without Fuze in Fiber Container PA 55.
9275181 - Packing and Marking for Box, Wirebound Packing, Ammunition for 105MM, Howitzer Ammunition in Fiber Container, M105A3.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Material and parts shall be in accordance with applicable drawings and specifications (see 6.10).

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.8).

3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9211611, 9216724 or 9212653, and with all requirements specified in applicable specifications.

3.3 Cavitation in high explosive charge.

3.3.1 Projected cavities. Cavities having a projected length of 1/32 inch (in.) or less shall be disregarded. Cavities within the explosive charge shall not exceed the requirements specified in Table I, when determined as specified in 4.5.1.

Table I. Cavitation.

	(Segments see Figure 1) A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.).	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks, and annular rings (in.).	1/8	1/2	1/2	3/4
Piping cavities, maximum (max.) Projected area (sq. in.).	0	1/4	1/2	-
Piping cavities, max. projected width (in.).	0	1/4	1/4	-
Cracks, max. projected width (in.).	1/32	1/32	1/32	-
Annular rings, max. projected width (in.).	0	0	1/4	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

*If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch.

Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudial axis (length) three times or greater than the width.

An Annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

3.3.2 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well permitted by the drawing, and provided no point extends above the flat surface on which the liner would rest. Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 9211610).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. Not more than two transverse cracks shall be permitted in any charge, and not more than one transverse crack shall be permitted in Segment A.

3.3.5 Base separation. There shall be no separation between the shell base and explosive charge.

3.3.6 Metal parts defects. There shall be no cracks, pipes or thin wall sections in the projectile.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge.

3.4 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

3.4.1 Muzzle velocity. The corrected muzzle velocity of the projectile at ambient temperature shall be not more than 1550 feet per second (473 meters per second) nor less than 1500 feet per second (457 meters per second) and the standard deviation of the lot shall be not greater than 8 feet per second (2.5 meters per second).

3.4.2 Premature bursts and security of parts. There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact.

3.4.3 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight when fired with propellant adjusted to give a pressure of 42,000 pounds per square inch (psi) plus or minus 1500 psi.

3.4.4 Functioning. The fuze shall cause the projectile to function high order (applicable only to cartridges with fuzed projectiles per dwg. 9216724 or 9212653).

3.5 Loading of high explosive

3.5.1 Mating surfaces. The mating surfaces of threads shall be free of caked explosives, and all threads shall be free of foreign material.

3.5.2 Solid explosives. Solid explosives shall not be present in the projectile before loading or be added to the shell during pouring or cooling.

3.5.3 Coating. Projectile having the interior wall imperfectly coated shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile.

3.5.4 Projectile interior cavity. The interior cavity of the projectile prior to loading shall be free from foreign matter.

3.5.5 Fuze well. The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.5.6 Pool of paint. There shall be no pool of paint in the bottom of the cavity prior to loading of the explosive.

3.5.7 Exterior surface. There shall be no explosive on the exterior surface of the projectile after loading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Cartridge. All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust and other foreign material. Care shall be exercised at all times so that the rotating band of each projectile will not be damaged by being nicked or burred.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of:

Part Description	Drawing	Quantity
Cartridge, 105MM, HE, M1, LAP	9211611, 9216724 or 9212653	42
Projectile, 105MM, HE, M1, Loading Assy	9211610	10

4.3.2 Inspection to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-45195E(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9211611, 9216724 or 9212653
	Cartridge, 105MM, HE, M1 Loading Assembling Packing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 105MM, HE, M1 Loading Assembling and Packing (Dwg 9211611, 9216724 or 9212653				
	Examination for defects	42(a)		3.1	4.4.2.9
	Proving Ground Test (Muzzle Velocity-Functioning	25(a)		3.4	4.4.3.3.1.1, 4.4.3.3.4.1, 4.4.3.3.4.2, 4.5.2
	Proving Ground Test (Safety)	15(a)		3.4	4.4.3.3.3.1, 4.5.2
	Examination for defects	b(a)		3.1	4.4.2.5 thru 4.4.2.8
	Projectile, 105MM, HE, M1 Loading Assembly (Dwg 9211610)				
	Examination for defects	10		3.1	4.4.2.3, 4.4.2.4
	Radiographic Inspection	10		3.3	4.5.1

NOTES: (a) Use same samples for these tests and examinations.
(b) Two cartridges shall be disassembled to allow checking of the examinations from para 4.4.2.5 thru 4.4.2.8.

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projectile shall contain:

a. Cartridge cases of one lot interfix number from one manufacturer.
b. Projectile metal parts of one lot interfix number from one manufacturer.
c. Propelling charges from not more than one lot.
d. Loaded primers of one lot interfix number from one manufacturer.
e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.
f. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.
g. TNT of the same type from one lot interfix number from one manufacturer; or Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.
h. Loaded projectile of same weight zone.
i. Loaded fuzes from one lot interfix number from one manufacturer.

4.4.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Projectile, Loading Assembly, prior to loading explosive	1 OF 1	9211610
			NEXT HIGHER ASSEMBLY: 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in charge cavity		100%	3.5.4	Visual
2	Pool of paint or blisters in projectile cavity		100%	3.5.6	Visual/Manual
Major					
101	Interior protective coating with bare spots		0.40%	3.5.3	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile, Loading Assembly, (prior to assembly of liner)	1 OF 1	9211610
			NEXT HIGHER ASSEMBLY: 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of explosive on exterior of shell body		100%	3.5.7	Visual
Major					
101	Depth of fuze well		0.40%	3.2	Gage
102	Foreign matter in fuze well		0.40%	3.5.5	Visual
Minor					
201	Diameter of fuze well, max.		0.65%	3.2	Gage
202	Surface cavitation excessive		0.65%	3.3.2	Visual
203	Presence of explosive on fuze threads		0.65%	3.5.1	Visual
204	Anti-seize compound not completely covering threads		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.0%	3.6	Visual

COMMENTS:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Projectile, Loading Assembly, (prior to assembling supplementary charge)		9211610
			NEXT HIGHER ASSEMBLY 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
Major					
101	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body		0.40%	3.2	Gage
102	Depth to bottom of liner, minimum		0.40%	3.2	Gage
Minor					
201	Liner damaged		0.65%	3.2	Visual
202	Foreign matter in liner cavity including explosive		0.65%	3.2	Visual
203	Liner removable by hand		0.65%	3.2	Manual
204	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.4	Projectile, Loading Assembly, (prior to assembly of closing plug)		1 OF 1	9211610
				NEXT HIGHER ASSEMBLY: 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pad on supplementary charge missing		100%	3.2	Visual
2	Supplementary charge missing		100%	3.2	Visual
3	Projectile zone weight incorrect by two or more zones		100%	3.2	Balance
Major					
101	Diameter of bourrelet max (applies if bourrelet has been retouched		100%	3.2	Gage
102	Weight of loaded shell incorrect by one zone with supplementary charge and liner but without a spacer and closing plug		0.40%	3.2	Balance
103	Supplementary charge not removable by hand		0.40%	3.2	Manual
104	Weight zone incorrect		0.40%	3.2	Visual
105	Supplementary charge inverted		0.40%	3.2	Visual
Minor					
201	Supplementary charge damaged		0.65%	3.2	Visual
202	Spacer missing or damaged		0.65%	3.2	Visual
203	Rotating band damaged		0.65%	3.2	Visual
204	Protective coating with total damaged area in excess of 1/4 square inch		0.65%	3.2	Visual
205	Marking missing, misleading or unidentifiable (when applicable)		0.65%	3.2	Visual
206	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8838130, 8838129
4.4.2.5	Primer (prior to assembly to case)				NEXT HIGHER ASSEMBLY 9211611, 9216724, 9212653
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	One or more flash holes missing		100%	3.2	Visual
Major	None defined				
Minor	None defined				

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Cartridge case (prior to assembling propelling charge)	1	1	8595386, 7548409, 7548025
				NEXT HIGHER ASSEMBLY 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Primer missing or damaged		0.40%	3.2	Visual
Minor 201	Foreign matter in interior of case		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.7	Propelling charge (prior to assembly to cartridge case)		1	1	9205472
					NEXT HIGHER ASSEMBLY: 9211611, 9216724, 9212653
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Increment missing or propellant missing		100%	3.2	Visual
Major 101	Any opening in bag large enough to permit escape of propellant		0.40%	3.2	Visual
Minor 201	Lacing cord broken		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-04-22T02:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Cartridge case with propelling charge (prior to assembling to projectile)	1	1	8595386
				NEXT HIGHER ASSEMBLY 9211611, 9216724, 9212653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect sequence or increments		100%	3.2	Visual
102	Foil side of bag, charge 5, not facing primer		0.40%	3.2	Visual
Minor	None defined				

DOTBD:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	Cartridge assembly	1	1	9211611, 9216724, 9212653
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer above flush		100%	3.2	Gage
Major					
101	Torque test failure of plug or fuze		0.40%	3.2	Test/4.5.3
102	Any component missing		0.40%	3.2	Visual
103	Staking of fuze missing or improper (when applicable)		0.40%	3.2	Visual
Minor					
201	Exterior coating of shell with an aggregate of bare spot in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking misleading, unidentifiable or missing including incorrect color (except zone weight)		0.65%	3.2	Visual
203	Plug or fuze not fully seated		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.10	Unsealed fiber container		1 1	7549073 or 9258026
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packing component missing		0.40%	3.2	Visual
Minor					
201	Cartridge cannot be removed by hand		0.65%	3.2	Manual
202	Container cap cannot be removed by hand		0.65%	3.2	Manual
203	Glue or asphalt on cartridge		0.65%	3.2	Visual
204	Packing components improperly assembled		0.65%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45195E(AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.11	Sealed fiber container	1 1	7549073 or 9258026
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container cut or damaged through all asphalt impregnated layers		0.40%	3.2	Visual-Manual
102	Metal end loose or distorted		0.40%	3.2	Visual-Manual
103	Tape incomplete or badly wrinkled		0.40%	3.2	Visual
Minor					
201	Tear tab length, min		0.65%	3.2	Gage
202	Contents loose		0.65%	3.2	Manual
203	Gap between cover and body container exceeding 1/8 inch		0.65%	3.2	Visual
204	Cuts, scuffs or gouges in outer layers		0.65%	3.2	Visual
205	Marking misleading, missing or unidentifiable		0.65%	3.2	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.12	Sealed wood packing box	1	1	7549072 or 9258027 9271850, 9275181
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Hardware or strapping missing, broken or loose		0.40%	3.2	Visual-Manual
103	DOD symbol missing or incorrect		0.40%	3.2	Visual
104	Board broken		0.40%	3.2	Visual
Minor					
201	Strapping or hardware improperly assembled		0.65%	3.2	Visual-Manual
202	Handle missing or insecure		0.65%	3.2	Visual-Manual
203	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
204	Marking misleading, missing or unidentifiable		0.65%	3.2	Visual
205	Contents loose		0.65%	3.2	Manual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and the following:

4.4.3.1 Check test for possible deterioration of primer, percussion, M28, (see specification MIL-P-46565). If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.5).

4.4.3.2 Cavitation (see 3.3, except 3.3.2). Metal parts defects and foreign material defects are as follows:

Critical:

1. Cavitation, cracks, or annular rings in Segment A, B, C or D in excess of that permitted in Table I.
2. Cracks in excess of that permitted in 3.3.4.
3. Base separation in excess of that permitted by 3.3.5.
4. Metal parts defects in excess of that permitted by 3.3.6 (see Section 6.11)
5. Foreign material in excess of that permitted by 3.3.7.

The sampling plan and procedures for this test shall be in accordance with MIL-STD-1235, using a CSP-1 plan having an AQL of .010 percent. The appropriate production interval shall be as specified by the procuring activity. The test shall be performed as specified in 4.5.1, using equipment in accordance with 4.4.4.

4.4.3.3 Proving ground.

4.4.3.3.1 Muzzle velocity (see 3.4.1) - Major defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.4.3.3.1.1 First three lots. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter I and AQL 1.0 percent. In addition, if the standard deviation of the sample projectile multiplied by the factor 0.75 exceeds the applicable lot requirement, the lot shall be rejected.

4.4.3.3.1.2 Production lots. After the three consecutive lots have met the criteria of 4.4.3.3.1.1, the sampling plan shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code Letter F and AQL of 1.0 percent. In addition, if the standard deviation of the sample projectile multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

4.4.3.3.2 Premature bursts and parts security. Premature bursts and security of parts (see 3.4.2) Critical defect. If any of the sample projectiles used in any of the tests fail to comply with the applicable requirements, the lot shall be rejected.

4.4.3.3.3 Safety. (see 3.4.3) - Critical defect.

4.4.3.3.3.1 First three lots. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, fifteen sample cartridges shall be randomly selected from each lot for this test. If any projectile fails to comply with the requirements, the lot shall be rejected.

4.4.3.3.3.2 Production lots. After three consecutive lots have met the criteria of 4.4.3.3.3.1, five sample cartridges shall be randomly selected from each lot for this test. If any projectile fails to comply with the requirements, the lot shall be rejected.

4.4.3.3.4 Functioning.

4.4.3.3.4.1 Fuze, M557. (see 3.4.4) Major defect.

4.4.3.3.4.1.1 First three lots. The cartridges used in 4.4.3.3.1.1 shall be used for this test. If three or more projectiles fail to function high order, the lot shall be rejected.

4.4.3.3.4.1.2 Production lots. After three consecutive lots have met the criteria of 4.4.3.3.4.1.1, the cartridges used in 4.4.3.3.1.2 shall be used for this test. If two or more projectiles fail to function high order, the lot shall be rejected.

4.4.3.3.4.2 Fuze, M564. (see 3.4.4) Major defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.4.3.3.4.2.1 First three lots. The cartridges used in 4.4.3.3.1.1 shall be used for this test. If four or more projectiles fail to function high order, the lot shall be rejected.

4.4.3.3.4.2.2 Production lots. After three consecutive lots have met the criteria of 4.4.3.3.4.2.1, the cartridges used in 4.4.3.3.1.2 shall be used for this test. If three or more projectiles fail to function high order, the lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the table starting in Paragraph 4.4.2.1. The contractor shall submit for approval of inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Determination of cavities. The sample projectiles shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 6.3). In addition, the sensitivity of the radiographic technique shall be 2.0 percent and minimum photographic density of the film used shall be not less than 1.25 at every point within the projected area of explosive, including the base and rotating band area, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.1 and 4.5.1.2. In the event an original radiograph leaves doubt as to its acceptability, the projectile shall be reradiographed 90 degrees from the original exposure. Acceptability shall be determined on the basis of the film giving the largest projected area of defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.5.1.1 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of the projectile so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectile and frequency of sensitivity check shall be no less than one in each fifty radiographs and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are sharply defined on the radiograph.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter see MIL-STD-453.

4.5.1.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities it is recommended that standards be used (6.8). Comparison of the film with standards may be made visually, however, the use of a densitometer is recommended.

4.5.1.3 Radiographic film. Only film type and vendor as listed herein are approved for use in radiographing the projectile. A substitute item shall not be used before prior testing and approval by the technical agency (see 6.12).

4.5.2 Proving ground. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired with M67 propelling charges in the M2A2 howitzer tube. Observation shall be made for conformance to all requirements. (see 6.6).

4.5.2.1 Muzzle velocity and functioning.

4.5.2.1.1 Without fuze. The sample cartridges shall be assembled with live or dummy fuzes and fired for muzzle velocity at zone 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.5.2.1.2 Fuze, M557. For lots with 25 sample cartridges, 15 of the fuzes shall be set "super quick" and 10 set "delay". For lots with 10 sample cartridges, 5 shall be set "super quick" and 5 "delay". The cartridges shall be fired at zone 7 with impact on firm ground.

4.5.2.1.3 Fuze, M564. All fuzes shall set for time. For lots with 25 sample cartridges, 15 fuzes shall be set for 90 seconds and 10 fuzes shall be set for 5 seconds. For lots with 10 sample cartridges, 5 fuzes shall be set for 90 seconds and 5 fuzes shall be set for 5 seconds. The cartridges shall be fired at zone 7 (with impact on firm ground when set for 90 seconds).

4.5.2.2 Safety. The sample cartridges shall be fired with propelling charges adjusted to give a pressure of 42,000 psi plus or minus 1500 psi in a weapon with the M103 tube or a ballistically equivalent tube. In addition, for the first three production lots from each manufacturer (loading facility), a smear camera shall be used to observe the projectile in flight in order to record any possible malfunctioning. (Cartridges furnished without fuzes shall be assembled with live or dummy fuzes for this test).

4.5.3 Torque tests. These tests shall be performed on the sample selected in accordance with 4.4.2.9. A thin line shall be drawn across the joint in question to aid in observing movement. The minimum disassembly torque specified on the drawing shall be applied to the fuze in the disassembly direction and the maximum disassembly torque specified on the drawing shall be applied to the plug in the disassembly direction.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Fiber container. The cartridges shall be preserved and packaged in accordance with dwg 7549073 or 9258026.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Level A. Fiber Container (dwg. 7549073) shall be packed in accordance with dwg. 7549072 or dwg. 9275181. Fiber Container (dwg. 9258026) shall be packed in accordance with dwg. 9271850 or dwg. 9258027 as directed by Government Procuring Activity.

5.3 Marking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

5.3.1 Marking. Marking of the wood box shall be in accordance with dwg. 7549072, 9258027 or 9271850. The fiber container shall be marked in accordance with dwg. 7549073 or 9258026.

6. NOTES

6.1 Intended use. The items covered by this specification are intended for use on the 105MM, HE, M1 projectile.

6.2 Ordering data. See MIL-A-48078(AR).

6.2.1 Data card. Data cards are required and shall be prepared. See MIL-A-48078(AR).

6.3 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAR-Q(D), Dover, New Jersey 07801-5001, for review and approval upon implementation of this Military specification and subsequently at each 6 month period of production.

6.4 Submission of inspection procedures and equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAR-I(D), Dover, New Jersey 07801-5001.

6.5 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.6 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commander
US Army Armament Research and Development Center
Dover, New Jersey 07801-5001
ATTN: AMSMC-QAR-Q(D)

6.7 Photographic densities. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York has been found satisfactory as a source of information of filter densities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

6.8 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.9 Standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ABCA-Army-STD-56B. When amendment, revision, or cancellation of this specification is proposed, the department custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.10 TNT frothing. When TNT shows definite frothing tendencies (which may cause production of unsatisfactory charges) 0.1 percent maximum of sorbitan trioleate in accordance with Specification JAN-S-547 may be used. Projectile lots loaded with TNT which has been so treated should have data cards annotated accordingly.

6.11 Metal parts defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.12 Radiographic film. List of approved radiographic film vendors are as follows:

Eastman Kodak Company
1187 Ridge Road, West Rochester
New York 14650

Film Types: AA, M, or T

E. I. Dupont deNemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-A621)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

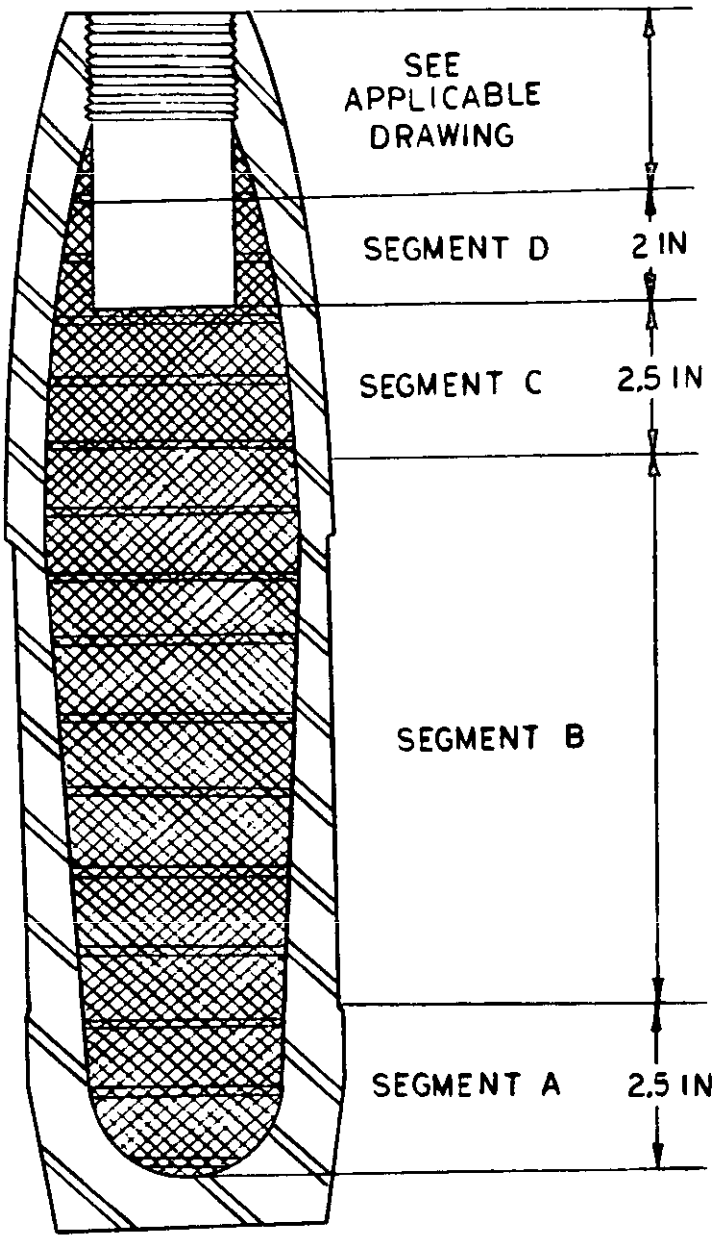
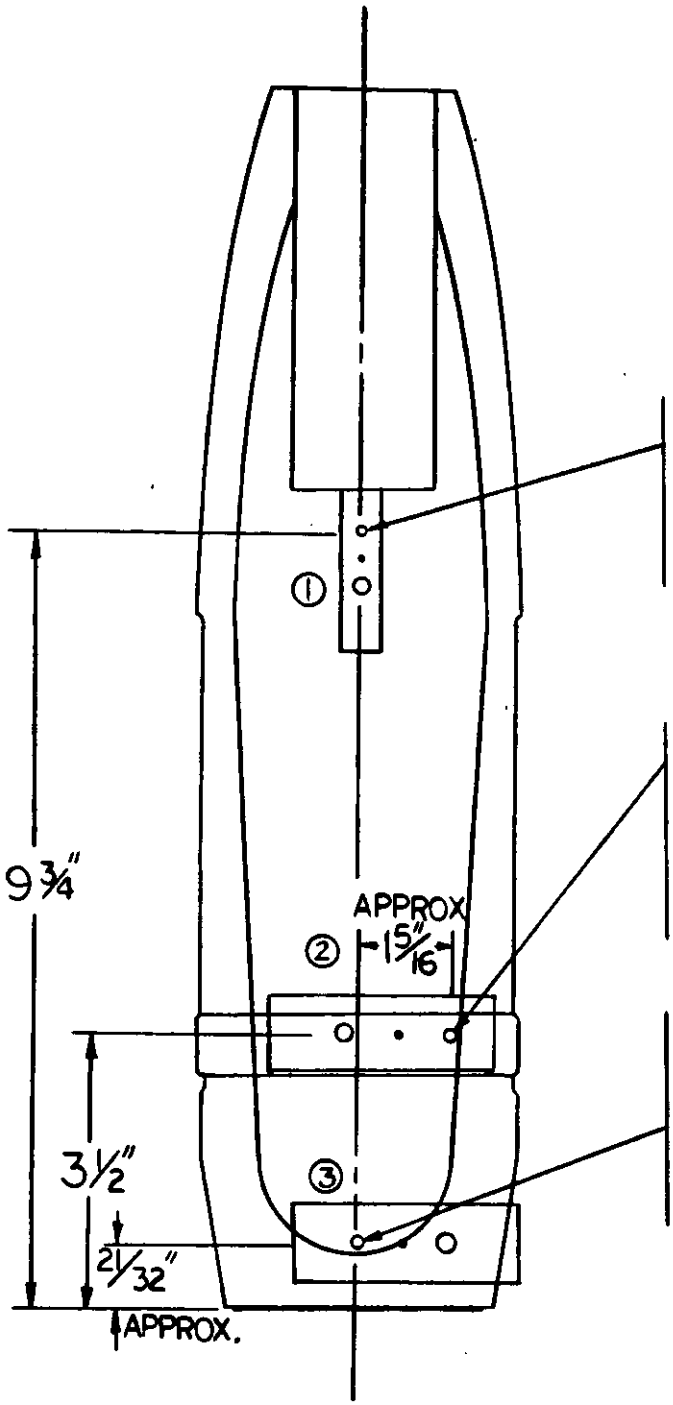


FIGURE 1. Cavitation in high explosive charge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.



Penetrameters Constructed Per MIL-STD-453
Dimensions shown are for location of center 2T holes.

PENETRAMETER FOR 1.5" STEEL
.75 Total steel wall plus 3.375" explosive.
1.5" equivalent thickness
2% penetrameter thickness = .030"
2T Hole diameter = .060"

PENETRAMETER FOR 2.7" STEEL
2.57" Total Steel wall plus .82" explosive at 1/16" from tangent point.
2.74" equivalent thickness
2% penetrameter thickness = .054"
2T Hole diameter = .108"

PENETRAMETER FOR 3.1" STEEL
3.0" Total steel wall plus .75" explosive at 1/16" from tangent point.
3.15" equivalent thickness
2% penetrameter thickness = .062"
2T Hole diameter = .124"

FIGURE 2. Projectile 105MM, HE, M1

Radiographic penetrameter requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Munitions &
Chemical Command
ATTN: AMSMC-QA
Dover, NJ 07801-5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-45195E (AR)	Cartridge, ±05MM, HE, M1 Loading, Assembling And Packing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil – Downloaded: 2026-10-29T02:33Z
Check the source to verify that this is the current version before use.